A Learning-Assisted LRT Method for Massive MIMO Radar Near-Field Detection Under Model Mismatch

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***Abstract*—This paper investigates near-field detection for massive multiple-input multiple-output (MIMO) radar with unknown parameters. The model-based likelihood ratio test (LRT) is optimal under the Neyman-Pearson criterion but hard to implement directly due to unknown parameters. In traditional signal processing, the generalized likelihood ratio test (GLRT) is widely used for detection with unknowns, but it degrades significantly under model mismatch when the assumed model differs from the actual environment. This work proposes a learning-assisted LRT detector that uses the LRT to form the test statistic and integrates learning networks to learn from data. By training on datasets that may include mismatched data to maximize detection performance, the learning-assisted LRT maintains good performance even under model mismatch.**

***Index Terms*—Massive multiple-input multiple-output (MIMO) radar, near-field detection, learning-assisted LRT, model mismatch.**

I. Introduction

Massive multiple-input multiple-output (MIMO) radar achieves significant performance gains with a large number of antennas [1], [2], [3]. Target detection is a core task for massive MIMO radar, but most existing work focuses on far-field signal model [4], [5], [6], [7], [8], [9], [10]. Near-field studies are limited and mostly address communication systems [11], [12], [13], [14], [15]. Among near-field radar works, most focus on parameter estimation [16], [17], [18], [19], [20], [21], with little attention to detection. This paper therefore considers near-field detection for massive MIMO radar.

In traditional signal processing, the likelihood ratio test (LRT) under the Neyman-Pearson (NP) criterion achieves optimal detection performance. However, radar detection often involves many unknown target and clutter parameters, making direct implementation difficult. The generalized likelihood ratio test (GLRT) is the most widely used method for detection with unknowns [22], but its performance depends on accurate environmental modeling. Under complex environments, model mismatch may occur [23], [24], causing significant GLRT performance degradation.

This paper studies massive MIMO radar near-field detection. Based on the NP criterion, we propose a learning-assisted LRT detector, where the LRT provides prior information and the learning networks enable the detector to learn from data. The performance of the learning-assisted LRT under model mismatch is analyzed. Simulation results demonstrate the effectiveness of the proposed method, especially under model mismatch, where the learning-assisted LRT suffers less performance degradation compared with purely model-based detectors.

II. Signal Model

Consider a passive massive MIMO radar which has $M \gg 1$ transmitters and $N \gg 1$ receivers. The target, if present, is placed at the position

$$\boldsymbol{\theta} = [x, y]^T. \tag{1}$$

The baseband equivalence of signal transmitted from mth transmitter can be written as $\sqrt{E_m} s_m(kT_s)$, where $k = 1, \ldots, K$ represents an index that varies across different snapshots, K the total number of time samples, E_m the transmitted energy, and T_s the sampling period. Here, a general near-field signal model with spherical wave is adopted, where the target is positioned at near field with respect to transmitter side and receiver side. Thus, the signal received by the nth antenna at time kT_s can be expressed as [18]

$$y_n[k] = s_n[k] + c_n[k], \tag{2}$$

where

$$\begin{aligned} s_n[k] = \sum_{m=1}^{M} \frac{\zeta_{nm}\sqrt{E_m \gamma_0 \eta_0}}{d_{t,m} d_{r,n}} \exp\left\{-j\frac{2\pi}{\lambda} d_{t,m}\right\} s_m(kT_s - \tau_{nm}) \\ \times \exp\left\{-j\frac{2\pi}{\lambda} d_{r,n}\right\}, \end{aligned} \tag{3}$$

ζ_{nm} is the unknown deterministic complex reflection coefficient for the (n, m)th path, λ is carrier wavelength, $d_{t,m}$ denotes distance from the mth transmit antenna to target, $d_{r,n}$ is distance from target to nth receiver, γ_0 represents complex gain at reference distance of 1m for transmitter side, η_0 is complex gain at reference distance of 1m for receiver side, τ_{nm} denotes delay for the (n, m)th path. The term $c_n[k] \sim \mathcal{CN}(0, \sigma^2)$ denotes zero-mean complex Gaussian clutter-plus-noise with unknown variance σ^2.

The received signal vector collecting K samples for the nth receive antenna is

$$\boldsymbol{y}_n = [y_n[1], \ldots, y_n[K]]^T = \boldsymbol{s}_n + \boldsymbol{c}_n, \tag{4}$$

where

$$\boldsymbol{s}_n = [s_n[1], \ldots, s_n[K]]^T, \tag{5}$$

and $\boldsymbol{c}_n = [c_n[1], \ldots, c_n[K]]^T$. Stack $\boldsymbol{y}_n$ from all receivers to obtain the received signal matrix

$$\mathbf{Y} = [\boldsymbol{y}_1, \ldots, \boldsymbol{y}_N] = \mathbf{S} + \mathbf{C}, \tag{6}$$

where $\mathbf{S} = [\boldsymbol{s}_1, \ldots, \boldsymbol{s}_N]$, and $\mathbf{C} = [\boldsymbol{c}_1, \ldots, \boldsymbol{c}_N]$.

Stacking all the unknown parameters into a vector, including the signal-related vectors, and the variance of clutter-plus-noise,

$$\boldsymbol{\phi} = [\boldsymbol{s}_1^T, \ldots, \boldsymbol{s}_N^T, \sigma^2]^T. \tag{7}$$

The near-field detection problem in passive massive MIMO radar is

$$\begin{aligned} H_0: \quad & \mathbf{Y} = \mathbf{C} \\ H_1: \quad & \mathbf{Y} = \mathbf{S} + \mathbf{C}, \end{aligned} \tag{8}$$

where H_0 represents the no target case, and H_1 denotes the target presence case.

III. Learning-Assisted LRT Detector

A. LRT for Near-Field Detection

The likelihood function under H_1 can be expressed as

$$p(\mathbf{Y}|H_1; \boldsymbol{\phi}) = \frac{1}{(\pi\sigma^2)^{NK}} \exp\left\{ -\sum_{n=1}^{N} \frac{(\boldsymbol{y}_n - \boldsymbol{s}_n)^H (\boldsymbol{y}_n - \boldsymbol{s}_n)}{\sigma^2} \right\}, \tag{9}$$

and the likelihood function under H_0 is $p(\mathbf{Y}|H_0; \sigma^2)$

$$p(\mathbf{Y}|H_0; \sigma^2) = \frac{1}{(\pi\sigma^2)^{NK}} \exp\left\{ -\sum_{n=1}^{N} \frac{\boldsymbol{y}_n^H \boldsymbol{y}_n}{\sigma^2} \right\}. \tag{10}$$

Based on the NP criterion, the LRT conditioned on unknown parameters $\boldsymbol{\phi}$ that solves near-field detection in (8) is

$$\begin{aligned} X_{LRT} &= \ln \frac{p(\mathbf{Y}|H_1; \boldsymbol{\phi})}{p(\mathbf{Y}|H_0; \sigma^2)} \\ &= \frac{1}{\sigma^2} \sum_{n=1}^{N} [2\Re\{\boldsymbol{y}_n^H \boldsymbol{s}_n\} - \boldsymbol{s}_n^H \boldsymbol{s}_n] \underset{H_0}{\overset{H_1}{\gtrless}} \gamma_{LRT}, \end{aligned} \tag{11}$$

where γ_{LRT} is detection threshold, and $\Re\{\cdot\}$ denotes real part.

The LRT in (11) achieves optimal detection performance but requires estimating the unknown $\boldsymbol{\phi}$ for implementation. The traditional model-based method for detection with unknowns is the GLRT, whose performance degrades significantly under model mismatch, i.e., the actual radar environment differs from the assumed model. Thus, we propose a learning-assisted LRT detector: it adopts the LRT-determined test statistic and integrates learning networks for data-driven capability. Trained on a dataset that may involve mismatched data to maximize detection performance, the proposed detector maintains good performance even under model mismatch.

The learning-assisted LRT detector has two modules: LRT module and decision module. The LRT module is used to obtain the test statistic and the decision module is employed for making detection decision. Multiplying both sides of equation (11) by σ^2 yields the following result

$$\tilde{X}_{LRT} = \sigma^2 X_{LRT} = \sum_{n=1}^{N} [2\Re\{\boldsymbol{y}_n^H \boldsymbol{s}_n\} - \boldsymbol{s}_n^H \boldsymbol{s}_n] \underset{H_0}{\overset{H_1}{\gtrless}} \sigma^2 \gamma_{LRT}. \tag{12}$$

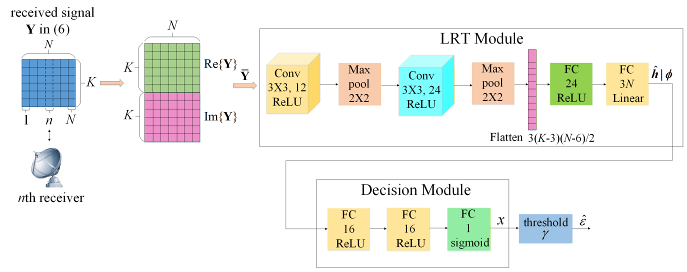

Fig. 1: Architecture of learning-assisted LRT detector.

The test statistic $\tilde{X}_{LRT}$ contains all valid information for optimal detection. As seen from (12), it consists of N summation terms, each being the sum of $2\Re\{\boldsymbol{y}_n^H \boldsymbol{s}_n\}$ and $\boldsymbol{s}_n^H \boldsymbol{s}_n$, indicating the test statistic can be derived via linear summation of these $2N$ terms. Thus, we only need to stack the $2N$ terms to form the test statistic vector. The test statistic conditioned on unknown vector $\boldsymbol{\phi}$ can be constructed as

$$\boldsymbol{h}|\boldsymbol{\phi} = [\boldsymbol{h}_1^T, \ldots, \boldsymbol{h}_N^T]^T, \tag{13}$$

where $\boldsymbol{h}_n = \left[2\Re\{\boldsymbol{y}_n^H \boldsymbol{s}_n\}, \boldsymbol{s}_n^H \boldsymbol{s}_n\right]^T$.

B. Learning-Assisted LRT Detector Architecture

Regarding the near-field detection in (8) as the classification problem, denote ε as detection indicator, where $\varepsilon = 0$ denotes H_0, and $\varepsilon = 1$ denotes H_1. Next, we develop the learning-assisted LRT detector to obtain the estimate of ε.

As shown in Figure 1, first, transform the complex matrix $\mathbf{Y}$ into a real matrix for further processing by taking its real and imaginary parts, such that,

$$f\colon \mathbf{Y} \in \mathbb{C}^{K \times N} \to \overline{\mathbf{Y}} \in \mathbb{R}^{2K \times N}, \tag{14}$$

where $\overline{\mathbf{Y}} = [\Re\{\mathbf{Y}\}; \Im\{\mathbf{Y}\}] \in \mathbb{R}^{2K \times N}$, and $\Im\{\cdot\}$ denotes the imaginary part.

1) LRT Module: LRT module aims to obtain estimate of $\boldsymbol{h}|\boldsymbol{\phi}$ in (13). The mapping of LRT module is written as follows

$$f_{\boldsymbol{\Theta}_{LRT}}\colon \overline{\mathbf{Y}} \in \mathbb{R}^{2K \times N} \to \hat{\boldsymbol{h}}|\boldsymbol{\phi} \in \mathbb{R}^{2N \times 1}, \tag{15}$$

where $\hat{\boldsymbol{h}}|\boldsymbol{\phi}$ is the estimate of $\boldsymbol{h}|\boldsymbol{\phi}$, and $\boldsymbol{\Theta}_{LRT}$ represents the trainable parameters of the LRT module.

As presented in Figure 1, the matrix $\overline{\mathbf{Y}}$ first passes through a 3×3 convolutional layer with 12 filters, followed by ReLU activation. A max pooling layer is then applied to reduce the data dimension, and the result is processed by another 3×3 convolutional layer and a second max pooling layer. After these operations, the output is flattened into a real vector with dimension $3(K-3)(N-6)/2$, which is fed into two fully connected layers: the first has 24 neurons with ReLU activation, and the second employs $2N$ neurons with linear activation, ultimately yielding $\hat{\boldsymbol{h}}|\boldsymbol{\phi}$.

2) Decision Module: Next, we feed $\hat{\boldsymbol{h}}|\boldsymbol{\phi}$ into the decision module to obtain the estimate of the detection indicator, whose mapping can be expressed as

$$f_{\boldsymbol{\Theta}_D}\colon \hat{\boldsymbol{h}}|\boldsymbol{\phi} \in \mathbb{R}^{2N \times 1} \to x, \tag{16}$$

where $\boldsymbol{\Theta}_D$ are trainable parameters of the decision module.

The decision module is a three-layer fully connected network. The first two layers each have 16 nodes with ReLU activation, and the third layer has a single neuron with sigmoid activation for target presence probability output. Detection is completed by comparing the module's output x with a threshold.

C. Two-Stage Supervised Training

The supervised learning is utilized to train the learning-assisted LRT detector. The training is divided into two phases. First, we perform preliminary optimization of the trainable parameters $\mathbf{\Theta}_{LRT}$ for the LRT module. In the second step, using the result of the preliminary optimization as the initial value for the $\mathbf{\Theta}_{LRT}$, we jointly optimize the trainable parameters for both the LRT module $\mathbf{\Theta}_{LRT}$ and the decision module $\mathbf{\Theta}_{D}$.

The mean squared error is used as loss for initial training of LRT module

$$\mathcal{L}_{\mathbf{\Theta}_{LRT}} = \frac{1}{S_{tr}} \sum_{i=1}^{S_{tr}} \|\boldsymbol{h}^{(i)} - \hat{\boldsymbol{h}}^{(i)}\|_2^2, \tag{17}$$

where S_{tr} is the number of training data, $\|\cdot\|_2$ denotes the L_2-distance, $\boldsymbol{h}^{(i)} = [(\boldsymbol{h}_1^{(i)})^T, \ldots, (\boldsymbol{h}_N^{(i)})^T]^T$ is the label, $\boldsymbol{h}_n^{(i)} = \left[2\Re\{(\boldsymbol{y}_n^{(i)})^H \boldsymbol{s}_n^{(i)}\}, (\boldsymbol{s}_n^{(i)})^H \boldsymbol{s}_n^{(i)}\right]^T$, $\boldsymbol{y}_n^{(i)}$ is received signal for the nth receiver, $\boldsymbol{s}_n^{(i)}$ denotes the signal-related part as shown in (5), and $\hat{\boldsymbol{h}}^{(i)}$ is the output of the LRT module. The labels $\boldsymbol{h}^{(i)}$ can be obtained because the training process can be artificially controlled and the related parameters of the transmitted signal and target are known. In the second stage, we jointly train the LRT module and decision module. The cross-entropy is used as loss

$$\mathcal{L}_{\mathbf{\Theta}_{LRT}, \mathbf{\Theta}_D} = -\frac{1}{S_{tr}} \sum_{i=1}^{S_{tr}} [\varepsilon^{(i)} \log(x^{(i)}) + (1 - \varepsilon^{(i)}) \log(1 - x^{(i)})], \tag{18}$$

where $\varepsilon^{(i)}$ is the label of target indicator, and $x^{(i)}$ is output.

IV. Discussion of Model Mismatch

Model mismatch, assumed model inconsistent with actual model, is common in practical radar environments and affects statistically modeled detectors, e.g., GLRT, learning-assisted LRT. For this work, the assumed model is given in (4), while the actual model retains the same signal-related components but with non-Gaussian clutter-plus-noise. Thus, the signal received by the nth receiver is

$$\tilde{\boldsymbol{y}}_n = \boldsymbol{s}_n + \tilde{\boldsymbol{c}}_n, \tag{19}$$

where $\tilde{\boldsymbol{c}}_n$ denotes actual received clutter-plus-noise, which follows the non-Gaussian distribution. Stacking $\tilde{\boldsymbol{y}}_n$ from all the receivers, the actual received signal matrix is

$$\tilde{\mathbf{Y}} = [\tilde{\boldsymbol{y}}_1, \ldots, \tilde{\boldsymbol{y}}_N] = \mathbf{S} + \tilde{\mathbf{C}}, \tag{20}$$

where $\tilde{\mathbf{C}} = [\tilde{\boldsymbol{c}}_1, \ldots, \tilde{\boldsymbol{c}}_N]$.

Considering model mismatch, the learning-assisted LRT is designed based on the assumed model (4) but trained and tested with data from the actual model (20). During the two-stage training, it optimizes trainable parameters using non-Gaussian assumption-generated datasets to maximize detection performance, enabling good performance even under model mismatch.

V. Numerical Examples

This section verifies effectiveness of the proposed learning-assisted LRT detector. For simplicity, consider the case with massive antennas at the receiver. Assume a passive massive MIMO radar with $M = 3$ transmitters, which are located at $(0, 0.0628)$m, $(0, 0)$m, and $(0, -0.0628)$m. For simplicity, set transmitting energy $E_m = E = 1$. The transmitting signal is a frequency spread single Gaussian pulse signal

$$s_m(kT_s) = \left(\frac{2}{W^2}\right)^{\frac{1}{4}} \exp\left(\frac{-\pi\,(kT_s)^2}{W^2} + j2\pi m\Delta f kT_s\right), \tag{21}$$

where $W = 0.01$s, $\Delta f = 500$Hz, and $T_s = 0.0001$s. The receiver side employs a uniform linear array with $N = 32$ antennas, in which the antenna n locates at $(100, (n - \frac{N+1}{2})g_r)$m, where $g_r = 0.0628$m is the antenna spacing. In simulations, we compare the detection probability of different algorithms while keeping false alarm probability below a preassigned level $P_{fa} = 0.001$. The number of time samples $K = 100$.

For the learning-assisted LRT detector, 20000 training, 6400 validation, and 10000 testing samples are generated per signal-to-clutter-plus-noise ratio (SCNR). Target positions in target-present samples are randomly selected from the near-field region, with no overlap between training, validation, and testing sets. Training parameters are listed in TABLE I.

Comparison methods include model-based LRT, GLRT, and data-aided GLRT. Simulations compare detector performance under model-matched and mismatched cases. For model-matched cases, the assumed and actual models both follow Gaussian distribution. For mismatched cases, the assumed model follows Gaussian distribution while the actual model follows non-Gaussian distribution. Detailed settings are provided in TABLE II.

In the first example, the compound Gaussian distribution is considered for the clutter-plus-noise in the actual model. For the nth receiver, the clutter-plus-noise vector is $\tilde{\boldsymbol{c}}_n = \sqrt{e_n}\boldsymbol{g}_n$ [26], where e_n is texture component, and $\boldsymbol{g}_n$ is K-dimensional speckle component, which follows zero-mean complex Gaussian distribution with covariance $\mathbf{\Sigma}_g$. The probability density function (PDF) of e_n is [26]

$$f\left(e_n\right) = \frac{1}{\Gamma(\theta_n)} \left(\frac{\theta_n}{\kappa_n}\right)^{\theta_n} e_n^{\theta_n - 1} \exp\left(-\frac{\theta_n}{\kappa_n} e_n\right), \tag{22}$$

where $\Gamma(\cdot)$ denotes Gamma function, θ_n represents shape parameter of e_n, and κ_n represents scale parameter of e_n. Set $\theta_n = \theta = 5$, $\kappa_n = \kappa = 0.2$. The covariance of $\boldsymbol{g}_n$ is $\mathbf{\Sigma}_g$, the (p, q)th element of which is $0.95^{|p-q|}$.

Figure 2 plots detection probability versus SCNR for the detectors in TABLE II under matched and mismatched conditions. Under model-matched case, the learning-assisted LRT and data-aided GLRT perform equivalently to the LRT and outperform the GLRT. Under mismatch, the learning-assisted LRT suffers the least performance loss, followed by the data-aided GLRT, while the GLRT degrades most severely. All detectors rely on statistical model and thus degrade under mismatch: the model-only GLRT is most sensitive. The data-aided GLRT uses extra data for parameter estimation but accumulates errors under mismatch. The learning-assisted LRT, however, is trained to account for mismatch effects and

TABLE I: Training Settings for Learning-Assisted LRT

Optimizer	Initialization	Epoch	Batch size	Data
Adam with default settings (learning rate=0.001, β_1=0.9, β_2=0.999, $\epsilon = 10^{-7}$)	Glorot uniform	20	64	training data: 20000 validation data: 6400 testing data: 10000

TABLE II: Descriptions of Detector Design under Matched and Mismatched Cases

Detector	Descriptions under matched case	Descriptions under mismatched case	Complexity (FLOPs)
LRT	LRT serves as benchmark for optimal detection performance, derived under Gaussian assumption with test statistic given in (12).	—	—
GLRT (derivation process is provided in the journal version.)	GLRT is derived under Gaussian assumption and tested under Gaussian clutter-plus-noise. Test statistic is $\frac{\lambda_1}{\sum_{n=1}^{N}\lambda_n}$, where $\lambda_1 > \lambda_2 > \ldots > \lambda_N$ are eigenvalues of matrix $\mathbf{Y}^H\mathbf{Y}$.	GLRT is derived under Gaussian assumption and tested under non-Gaussian clutter-plus-noise. Test statistic is $\frac{\tilde{\lambda}_1}{\sum_{n=1}^{N}\tilde{\lambda}_n}$, where $\tilde{\lambda}_1 > \tilde{\lambda}_2 > \ldots > \tilde{\lambda}_N$ are eigenvalues of matrix $\tilde{\mathbf{Y}}^H\tilde{\mathbf{Y}}$.	$\mathcal{O}(N^3)$
data-aided GLRT [25]	Data-aided GLRT is derived under Gaussian assumption. It first estimates signal-related vectors $\boldsymbol{s}_n$ from 10000 Gaussian-distributed data samples via MLE, then substitutes estimated $\boldsymbol{s}_n$ into LRT (12) to yield test statistic. This detector is tested under Gaussian clutter-plus-noise.	Data-aided GLRT is derived under Gaussian assumption. It first estimates signal-related vectors $\boldsymbol{s}_n$ from 10000 non-Gaussian-distributed data samples via MLE, then substitutes estimated $\boldsymbol{s}_n$ into LRT (12) to yield test statistic. This detector is tested under non-Gaussian clutter-plus-noise.	$\mathcal{O}(KN)$
learning-assisted LRT	Learning-assisted LRT is designed under Gaussian assumption, and is both trained and tested under Gaussian clutter-plus-noise.	Learning-assisted LRT is designed under Gaussian assumption, and is both trained and tested under non-Gaussian clutter-plus-noise.	$\mathcal{O}(KN)$

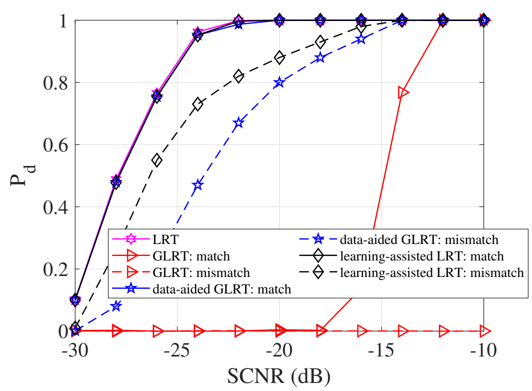

Fig. 2: Detection probability vs. SCNR under model match/mismatch, compound Gaussian distribution is used for actual model under mismatch, $N = 32$, $K = 100$, $P_{fa} = 0.001$.

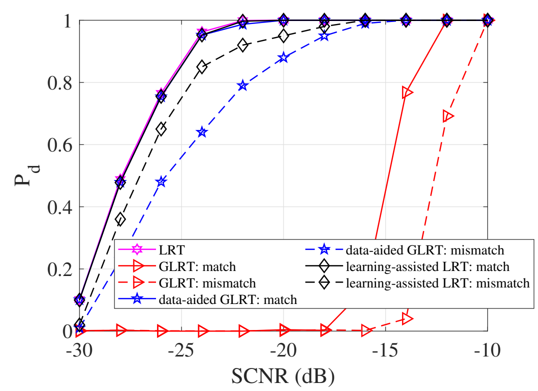

Fig. 3: Detection probability vs. SCNR under model match/mismatch, mixture Gaussian distribution is used for actual model under mismatch, $N = 32$, $K = 100$, $P_{fa} = 0.001$.

optimize detection performance, learning robust parameters that maintain effectiveness under mismatch. Computational complexity, measured in FLOPs (floating point operations), is listed in TABLE II.

In the second example, the clutter-plus-noise in actual model obeys mixture Gaussian distribution. The $c_n[k]$ is spatially and temporally white with PDF [27]

$$f(c_n[k]) = \omega f_0(c_n[k]; \sigma_0^2) + (1-\omega) f_1(c_n[k]; \sigma_1^2), \quad (23)$$

where ω denotes mixing parameter, $f_0(c_n[k]; \sigma_0^2)$ denotes PDF of zero-mean complex Gaussian distribution with variance σ_0^2, and $f_1(c_n[k]; \sigma_1^2)$ represents PDF of zero-mean complex Gaussian distribution with variance σ_1^2. Set $\omega = 0.1$, $\sigma_0^2 = 1$, and $\sigma_1^2 = 2$.

Figure 3 shows detection probability vs. SCNR for LRT, learning-assisted LRT, data-aided GLRT, and GLRT under matched and mismatched scenarios. Similarly, under matched case, the learning-assisted LRT and data-aided GLRT perform close to the optimal LRT. Under mismatch, the learning-assisted LRT incurs the least performance loss.

VI. Conclusion

This paper investigates near-field detection for massive MIMO radar and proposes a learning-assisted LRT detector based on the NP criterion. The detector includes an LRT module for test statistic construction and a decision module for decision-making. The performance under model mismatch is analyzed. Simulations validate effectiveness of the proposed method under both matched and mismatched conditions.

REFERENCES

[1] G. Xue, H. Li, and R. Liang, "Convergence-enhanced 1-bit DACs precoding for massive MIMO-OFDM systems via Anderson acceleration," *Digital Signal Processing*, vol. 168, p. 105545, 2026.

[2] N. Sadeghi, M. Azghani, and S. A. Mortazavi, "Joint pilot optimization and channel estimation using deep learning in massive MIMO systems," *Digital Signal Processing*, vol. 165, p. 105287, 2025.

[3] E. Björnson, L. Sanguinetti, H. Wymeersch, J. Hoydis, and T. L. Marzetta, "Massive MIMO is a reality—What is next?" *Digital Signal Processing*, vol. 94, pp. 3–20, Nov. 2019.

[4] M. Deng, L. Zhou, S. Zhang, Z. Wang, and Z. Cheng, "On the performance of one-bit matched filtering in massive MIMO radar," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 61, no. 1, pp. 208–222, Feb. 2025.

[5] Q. Lin, H. Shen, Z. Li, W. Xu, C. Zhao, and X. You, "One-bit transceiver optimization for mmwave integrated sensing and communication systems," *IEEE Transactions on Communications*, vol. 73, no. 2, pp. 800–816, Feb. 2025.

[6] M. Elfiatoure, H. Q. Ngo, and M. Matthaiou, "Coexistence between massive MIMO and radar communications: Performance analysis," *Journal of Communications and Information Networks*, vol. 8, no. 1, pp. 37–47, Mar. 2023.

[7] F. Rosu, "Dimension compressed CFAR for massive MIMO radar," *IEEE Geoscience and Remote Sensing Letters*, vol. 20, pp. 1–5, 2023.

[8] E. Krijestorac, R. Li, D. Cabric, J. McGraw, P. Powers, J. A. Vitaz, M. Salpukas, D. Eustice, and T. Allen, "Machine learning-assisted computationally efficient target detection and tracking in massive fully digital phased arrays," *IEEE Transactions on Radar Systems*, vol. 1, pp. 353–367, 2023.

[9] Z. Behdad, Ö. T. Demir, K. W. Sung, E. Björnson, and C. Cavdar, "Multi-static target detection and power allocation for integrated sensing and communication in cell-free massive MIMO," *IEEE Transactions on Wireless Communications*, vol. 23, no. 9, pp. 11 580–11 596, Sep. 2024.

[10] W. Zhai, X. Wang, X. Cao, M. S. Greco, and F. Gini, "Reinforcement learning based dual-functional massive MIMO systems for multi-target detection and communications," *IEEE Transactions on Signal Processing*, vol. 71, pp. 741–755, 2023.

[11] X. Cao, A. Chen, H. Yin, and L. Chen, "A low-complexity near-Field localization method based on electric field model," *IEEE Communications Letters*, vol. 28, no. 2, pp. 288–292, Feb. 2024.

[12] H. Lu, Y. Zeng, C. You, Y. Han, J. Zhang, Z. Wang, Z. Dong, S. Jin, C.-X. Wang, T. Jiang, X. You, and R. Zhang, "A tutorial on near-field XL-MIMO communications toward 6G," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 4, pp. 2213–2257, 2024.

[13] P.-Z. Xu, W. Xiang, and Q.-Y. Yu, "DFrFT-based near-Field channel estimation for extremely large-scale MIMO systems," *IEEE Communications Letters*, vol. 28, no. 8, pp. 1914–1918, Aug. 2024.

[14] Y. Wang, S. Sun, and C. Han, "Far- and near-Field channel measurements and characterization in the terahertz band using a virtual antenna array," *IEEE Communications Letters*, vol. 28, no. 5, pp. 1186–1190, May 2024.

[15] B. Qin, F. Zhou, W. Zhang, C. You, and B. Zheng, "Fast distance sampling for uniform circular array in near-field 3D beam-focusing," *IEEE Signal Processing Letters*, vol. 31, pp. 2210–2214, 2024.

[16] Z. Wang, X. Mu, and Y. Liu, "Near-field velocity sensing and predictive beamforming," *IEEE Transactions on Vehicular Technology*, vol. 74, no. 1, pp. 1806–1810, Jan. 2025.

[17] H. Wang, Z. Xiao, and Y. Zeng, "Cramér-Rao bounds for near-Field sensing with extremely large-scale MIMO," *IEEE Transactions on Signal Processing*, vol. 72, pp. 701–717, 2024.

[18] H. Hua, J. Xu, and Y. C. Eldar, "Near-field 3D localization via MIMO radar: Cramér-Rao bound analysis and estimator design," *IEEE Transactions on Signal Processing*, vol. 72, pp. 3879–3895, 2024.

[19] S. K. Yadav, S. R. M. Prasanna, and N. V. George, "Near-field source localization and beamforming in the spherical sector harmonics domain," *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 4, pp. 546–560, May 2024.

[20] J. Yu, Y. Xu, S. Zhang, and Z. Liu, "Symmetric interantenna spacing constrained sparse arrays for DOA estimation of mixed near-field and far-field source signals," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 60, no. 4, pp. 4679–4698, Aug. 2024.

[21] G. Wang, Y. Xiao, K. C. Ho, and L. Huang, "Unified near-field and far-field TDOA source localization without the knowledge of signal propagation speed," *IEEE Transactions on Communications*, vol. 72, no. 4, pp. 2166–2181, Apr. 2024.

[22] S. M. Kay, *Fundamentals of Statistical Signal Processing*, ser. Prentice Hall Signal Processing Series. Englewood Cliffs, N.J: Prentice-Hall PTR, 1993.

[23] K.-W. Peng, "Risk-aware low-pass importance sampling for graph signals under heterogeneous noise and model mismatch," *IEEE Signal Processing Letters*, vol. 33, pp. 556–558, 2026.

[24] A. M. Kuzminskiy, A. Elzanaty, G. Gradoni, F. Wang, and R. Tafazolli, "Near-field localization with physics-compliant electromagnetic model: Algorithms and model mismatch analysis," *IEEE Internet of Things Journal*, vol. 12, no. 15, pp. 30 533–30 547, Aug. 2025.

[25] B. Huang, D. Orlando, W.-Q. Wang, J. Jian, Y. Jia, W. Jia, and W. Liu, "GLRT-based adaptive target detection for FDA-MIMO radar in mainlobe deceptive jamming," *IEEE Sensors Journal*, vol. 25, no. 8, pp. 13 330–13 343, Apr. 2025.

[26] S. Xu, Y. Hao, Z. Wang, and J. Xue, "Persymmetric adaptive polarimetric detection of subspace range-spread targets in compound Gaussian sea clutter," *Journal of Systems Engineering and Electronics*, vol. 35, no. 1, pp. 31–42, Feb. 2024.

[27] E. J. Wegman, J. B. Thomas, and S. C. Schwartz, *Topics in Non-Gaussian Signal Processing*. Berlin, Heidelberg: Springer-Verlag, Oct. 1988.